

Kevin Nam

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About Me

I am currently pursuing a combined MS/PhD in the Security Optimization Research Lab at Seoul National University. I hold a Bachelor of Science in Electrical and Computer Engineering from the same university, with a focus on hardware/system/AI security and privacy. But my research fields are not limited to them. Back when I was an undergraduate student, I had been deeply passionate about hardware design and computer architecture. My first research topics were naturally related to efficient system integration (including hardware designs) for high-performance domain-specific computation. My research interests have expanded since then, including security and privacy-preserving computation. My current focused research topics are homomorphic encryption, multi-party computation, and trusted execution environments. As I deepen my expertise in these areas, I remain open to exploring new challenges and interdisciplinary fields within broader scopes.

Education

Seoul National University, BS in Electrical and Computer Engineering Mar 2014 – Feb 2020

- Left for military service for 2017-2019

Seoul National University, MS/Ph.D in Electrical and Computer Engineering Mar 2020 – Feb 2026 (expected)

Publications

- "Affinity-based Optimizations for TFHE on Processing-in-DRAM". (To Appear) *In the 30th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25)*, April 2025
Kevin Nam, Heonhui Jung, Hyunyoung Oh, and Yunheung Paek
- "LOHEN: Layer-wise Optimizations for Neural Network Inferences over Encrypted Data with High Performance or Accuracy". (To Appear) *In USENIX Security Symposium (Security'25)*, August 2025
Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek
- "An Efficient Hardware/Software Co-design for FALCON on Low-End Embedded Systems". *In IEEE Access*, 2024
Yongseok Lee, Jonghee Youn, Kevin Nam, Heon Hui Jung, Myunghyun Cho, Jimyung Na, Jong-Yeon Park, Seungsu Jeon, Bo Gyeong Kang, Hyunyoung Oh, Yunheung Paek
- "Accelerating N-bit Operations over TFHE on Commodity CPU-FPGA". *In IEEE/ACM International Conference on Computer-Aided Design (ICCAD'22)*, November 2022
Kevin Nam, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek
- "MeetGo: A trusted execution environment for remote applications on FPGA". *In IEEE Access*, 2021
Hyunyoung Oh, Kevin Nam, Seongil Jeon, and Yeongpil Cho, Yunheung Paek

Research Experience (w. fundings)

Zero-Error Privacy-Preserving Neural Networks funded by NRF, South Korea Oct 2023 – present

- Research on desining efficient and accurate neural network services over encrypted data
- Integrate non-mathematical methods (e.g., compiler) with the mathematical cryptosystems
- Currently participating as **the project leader**

Secure FHE Key Management System funded by CryptoLab, South Korea Mar 2023 – April 2024

- Designed TEE-based FHE key management system
- Developed the protocol to publish and deprecate keys to users specifically adapted for FHE
- Participated as the TEE software engineer

Neural Network Program to FHE Transpiler funded by ETRI, South Korea Oct 2023 – Nov 2023

- Designed a transpiler that changes a tensorflow, Pytorch program to FHE programs

- Developed the FHE routines that corresponds to the kernels used in tensorflow (e.g., convolution, activations)
- **Solely participated in the whole project**

PEC Platform funded by the National Intelligence Service, South Korea

Mar 2022 – Nov 2023

- Designed a Privacy-Enhancing-Computation (PEC) platform that exploits the use of various privacy-preserving techniques (e.g., FHE, MPC, TEE)
- Developed a framework that transpiles a normal code into privacy-preserving one, suggesting appropriate privacy-preserving technique for each partitions
- Participated as **the project leader** (designing the dataflow of the platform/framework)

FPGA Accelerator for CKKS funded by CryptoLab, South Korea

Mar 2022 – Nov 2022

- Designed an FPGA-based CKKS Accelerator for its bootstrapping algorithm
- Participated as **the algorithm analyzer** - designed the overall operation mapping and pipeline

FPGA Accelerator for FHE funded by the National Intelligence Service

Mar 2021 – Nov 2021

- Designed an FPGA-based TFHE accelerator exploiting the double parallelism of TFHE
- Developed a methodology to find the optimal setup of parallelism considering the system's memory bandwidth
- **Solely participated in the whole project**, from algorithm analysis to system integration

Honors & Awards

Undang Academic Award from KIPS, Best Graduate Student Paper Award

Dec 2021

NIPA Director Award from ASK 2023

Jun 2023

Award for Excellence in Teaching Assistant, from Seoul National University

Jan 2024

Logic circuit Design Course

Scholarship

BK 21+ Scholarship by the Ministry of Education of Korea

Mar 2020 - present

SNU graduate student scholarship by Seoul National University

Mar 2022 - Feb 2024

Activities

EuroSys'25 Shadow Program Committee

2025

Teaching (Assistant) Experience

Seoul National University

Logic Circuit Design Course

Sep 2023 - Dec 2023

Data Security & Privacy

Sep 2024 - Dec 2024

Skills

Languages: English, French, Korean

Programming Languages: C/C++, Python, HDL, JAVA

Tools/Frameworks: Vitis, Vivado, Xcellium, Synopsys DC, Tensorflow, Pytorch, Docker, Git, Linux

Privacy-Preserving Tools: MS SEAL, OpenFHE, HeaaN, TFHE-rs, OpenCheetah, and much more